

Two coupled ODEs

integrated jointly from $t = 0$ to $t = 1$

Pretrained flow matching model

$$v_{\theta}(x, t)$$

Primal · samples

$$\dot{x}_t = v_{\theta} - J_g^{\top}(\lambda_t + cg)$$

λ_t

$g(x_t)$

Dual · Lagrange multipliers

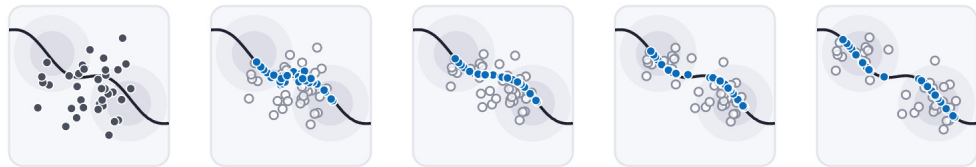
$$\dot{\lambda}_t = g(x_t)/(1-t)^p$$

Constrained samples

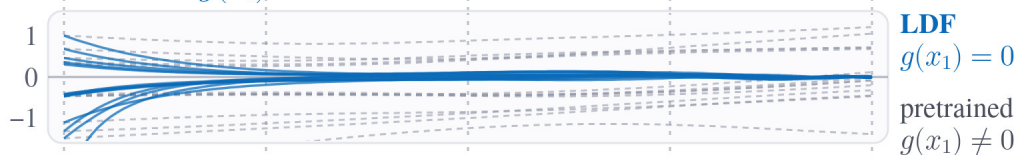
$$g(x_1) = 0$$

Primal samples x_t

— $g(x) = 0$ ○ pretrained ● LDF



Primal residual $g(x_t)$



Dual Lagrange multiplier λ_t

